The Accountability Gap.

A Microsoft Fabric case study in defensive T-SQL, hierarchy resolution and SCD Type 2 history in a Fabric Warehouse, for a forty-two billion dollar asset manager rebuilding its regulatory attribution chain.

01 The Recommendation
02 Context and Diagnosis
03 The Target
04 The Mandate
05 Acceptance
06 Discussion, Lab and Challenges

Veritas Global
Asset management, London, New York and Singapore

Microsoft Fabric

SECTION 01 · THE RECOMMENDATION

Close the accountability gap before the next regulatory cycle.

Three regulators require Veritas Global to name the Relationship Manager who was responsible for every flagged position at the moment it was opened. The roster table is overwritten on every personnel change, so that history is gone and the firm cannot answer the question.

NOT A REPORTING PROBLEM

Regulators do not accept current-state data in place of historical accountability. A firm that cannot rebuild its accountability chain at a point in time faces adverse findings, penalties and, in serious cases, limits on its licence.

THE AUTHORISATION

Leadership has authorised the data engineering team to design and deliver the T-SQL architecture that closes the gap permanently, before the next regulatory cycle. Four deliverables, each addressing one failure mode.

$2M to $5M
Regulatory exposure per audit cycle from an FCA, SEC or MAS adverse finding

15 hrs
Manual reconstruction per regulatory cycle, from memory and email

1 finding
Enough to trigger a mandate review across the whole institutional client base

SECTION 02 · CONTEXT AND DIAGNOSIS

Three regulators. One unanswerable question.

Veritas Global is a London-based institutional asset manager with offices in New York and Singapore. Founded in 2001, it manages forty-two billion dollars across equities, fixed income, derivatives and FX for pension funds, sovereign wealth funds and university endowments.

EXHIBIT 01 · REGULATORY LANDSCAPE

REGULATOR	JURISDICTION	KEY OBLIGATION
FCA	United Kingdom	Point-in-time accountability for every flagged position.
SEC	United States	Historical attribution records kept for at least seven years.
MAS	Singapore	A complete audit trail for every position above the regulatory threshold.

> *The regulator asked us a straightforward question. Who was the Relationship Manager responsible for this position on the fourteenth of October? We could not answer it, because we had overwritten the record when the team restructured in November.*
>
> OLIVER HARTWELL · MANAGING DIRECTOR, GLOBAL HEAD OF RISK

The failure surfaced in an FCA review in Q4 2025. The reviewer asked for point-in-time attribution of three positions opened in Q3. Two were managed by RMs who had since changed desk or title, and the roster query returned only their current state.

EXHIBIT 02 · THREE COMPOUNDING FAILURES

FAILURE	WHAT HAPPENS	MEASURE
DIRTY SOURCE DATA	**Notional values stored as text with embedded symbols. Dates stored as text in mixed formats. A plain cast fails the whole batch.**	100 percent of a batch lost on one bad cast
HIERARCHY BLINDNESS	**The reporting line runs five levels, from Analyst to Managing Partner. A single join against a flat roster cannot resolve it.**	5 levels in the accountability chain
ROSTER OVERWRITE	**When an RM changes title, manager or jurisdiction, the record is updated in place and the prior state is destroyed.**	0 historical states kept

WHY THE PAST CANNOT BE RECOVERED

History tracking can start today, and every future roster change will be kept. States already overwritten cannot be rebuilt. Oliver Hartwell and Priya Anand must hear this before go-live: the go-live date is the earliest date for reliable attribution.

WHY THE FUTURE MUST BE PROTECTED

After go-live every roster change becomes a new historical version, and the chain for any position opened after that date can be rebuilt exactly. Telling regulators what the system can and cannot do is itself a sign of governance maturity.

SECTION 03 · THE TARGET

Five levels. Three structural changes.

Each RM sits on one desk (Equities, Fixed Income or Multi-Asset) in one jurisdiction. Promotions, desk moves and jurisdiction transfers happen monthly, and until now each was written over the old record.

EXHIBIT 03 · RELATIONSHIP MANAGER HIERARCHY

TITLE	REPORTS TO	REGULATORY SIGNIFICANCE	DEPTH
Managing Partner	No manager	Ultimate accountability for every position in the firm.	1
Managing Director	Managing Partner	Senior accountability for desk-level exposure in a jurisdiction.	2
Director	Managing Director	Operational accountability for team positions and escalation.	3
Associate	Director	Day-to-day position management and direct client ownership.	4
Analyst	Associate	Position monitoring, first flags, escalation to the Associate.	5

Hierarchy_Depth counts the people in an RM's chain, from the RM up to the Managing Partner. An Analyst has depth 5 and the Managing Partner depth 1.

EXHIBIT 04 · OPERATING MODEL SHIFT

DIMENSION	CURRENT STATE	TARGET STATE
DATA QUALITY	**Dirty values crash the pipeline. Batches drop silently.**	A governed view absorbs every source failure. No position is lost.
NULL HANDLING	**Positions with no RM drop out of joins.**	Every position appears, with no RM shown explicitly.
EXPOSURE	**No classification. Analysts judge by hand.**	Every position carries an exposure category.
HIERARCHY	**The five-level chain cannot be resolved.**	The full chain from Analyst to Managing Partner in one query.
HISTORY	**Roster updates overwrite the prior state.**	Every roster change kept as a historical record.

SECTION 04 · THE MANDATE

Four deliverables. Each closes one failure mode.

You are the Lead Data Engineer, reporting to Priya Anand, Head of Regulatory Reporting. Oliver Hartwell's directive: close the accountability gap before the next regulatory cycle. Everything is built in the Fabric Warehouse `Warehouse_Veritas`.

EXHIBIT 05 · TABLES AND VIEWS

OBJECT	LAYER	DESCRIPTION
`stg.Flagged_Positions`	Staging	Nightly export from the legacy execution system, loaded from stg_Flagged_Positions.csv. Dirty values, malformed dates, text notionals.
`ref.Relationship_Managers`	Reference	The RM roster, loaded from ref_Relationship_Managers.csv. Flat today, SCD Type 2 after Deliverable 4.
`stg.New_RM_Roster`	Staging	Incoming roster changes for Deliverable 4. Created by the setup script in the lab.
`gold.Clean_Positions`	Gold	The governed view from Deliverable 1: clean, classified and attributed positions for regulatory reporting.

DELIVERY ROADMAP

DELIVERABLE	BUSINESS OUTCOME	CLOSES
D1 · gold.Clean_Positions	**Every flagged position visible in clean, classified, attributed form, whatever the source quality.**	Dirty data
D2 · RM leaderboard	**Which RMs resolve positions with regulatory sign-off, ranked against peers at the same title.**	Visibility
D3 · Hierarchy flattener	**The complete five-level chain for every RM.**	Hierarchy
D4 · SCD Type 2 MERGE	**Every future roster change kept as a historical record. Point-in-time attribution from go-live.**	Overwrite

EXHIBIT 06 · GOLD.CLEAN_POSITIONS

COLUMN	TYPE	BUSINESS PURPOSE	SOURCE
`PositionID`	VARCHAR	Unique identifier for each flagged position.	stg
`RMID / RM_Name`	INT / VARCHAR	RM identifier and name. Where no RM matches, RMID is NULL and RM_Name says so explicitly.	ref
`Asset_Class`	VARCHAR	Equity, Fixed Income, Derivative or FX.	stg
`Currency / Flag_Type`	VARCHAR	USD, GBP or SGD. Regulatory, Internal or Counterparty.	stg
`Status`	VARCHAR	Open, Resolved or Escalated.	stg
`Regulatory_Signoff`	VARCHAR	Y or N.	stg
`Clean_Settlement_Date`	DATE	Settlement date after a safe cast. 1900-01-01 where the source cannot be parsed.	Derived
`Clean_Flag_Date`	DATE	Flag date after a safe cast. 1900-01-01 where the source cannot be parsed.	Derived

COLUMN	TYPE	BUSINESS PURPOSE	SOURCE
`Clean_Notional_Value`	DECIMAL(20,2)	Notional after a safe cast, with no symbol or comma stripping. NULL where the value cannot be parsed.	Derived
`Exposure_Category`	VARCHAR	The band from Exhibit 07. Every row gets one.	Derived

Every numeric and date column from staging uses TRY_CAST or TRY_CONVERT, which return NULL instead of raising an error. One malformed row can no longer take the view offline. In return, every downstream query must handle NULLs and the sentinel date.

EXHIBIT 07 · EXPOSURE BANDS

EXPOSURE_CATEGORY	CLEAN_NOTIONAL_VALUE	NOTE
LOW EXPOSURE	Under 1,000,000	A safe cast of the source value.
MEDIUM EXPOSURE	1,000,000 to 9,999,999.99	
HIGH EXPOSURE	10,000,000 to 99,999,999.99	
CRITICAL EXPOSURE	100,000,000 and over	The band both challenges use.
UNCLASSIFIED	NULL: the value could not be parsed	Explicitly classified, never dropped.

SECTION 05 · ACCEPTANCE

Seven criteria. One honest boundary.

Each criterion maps to a deliverable and the failure mode it closes.

01	**Safe cast · D1**	Every dirty source value converts without an error. No position is lost to a malformed date or notional.
02	**Exposure · D1**	Every row carries one of the five Exhibit 07 bands. Unparseable notionals are Unclassified, never excluded.
03	**Null safety · D1**	No row is lost for a missing RM. Every position appears, with an explicit RM_Name where no RM is assigned.
04	**Leaderboard · D2**	Rankings are partitioned by Title. Only Resolved positions with sign-off Y count, and equal counts share a rank without skipping the next. Every RM with at least one qualifying position appears.
05	**Hierarchy · D3**	Every RM appears once, with full name, Title, Hierarchy_Depth as defined under Exhibit 03 and the complete Accountability_Chain. Missing levels are skipped, not shown as gaps.

06	**SCD Type 2 · D4**	A change to Title, ManagerID or Jurisdiction closes the prior record before the new one is active. Desk alone does not create a version. No historical row is overwritten.
07	**Historical integrity · D4**	After a change, the RM has two rows: one expired with an end date, one active.

The honest boundary. History tracking protects every roster change from go-live onward. Record the go-live date as the earliest date for reliable point-in-time attribution, and tell every regulator before they ask a question that crosses it.

SECTION 06 · DISCUSSION, LAB AND CHALLENGES

Five questions. Four deliverables. Two challenges.

Prepare a clear, defensible answer to each question. They test the reasoning behind your design, not the steps you followed.

Q1 · Safe casting	Your view uses TRY_CONVERT rather than CONVERT. What exactly happens when a value cannot be converted? What happens to the view if plain CONVERT meets one unparseable notional?
Q2 · The date sentinel	Your view puts 1900-01-01 in every date that cannot be parsed. What does that mean for any downstream query that filters or windows on the clean dates, and what guard must each one include?
Q3 · Deduplication before joining	Your hierarchy flattener deduplicates the roster before joining. Why? What failure does it prevent, and what would the output look like without it?
Q4 · The routing pattern	Your SCD Type 2 MERGE routes source rows to different clauses using a derived column. How does the routing work, and why can one source row not trigger both clauses even when the same RM appears in both branches?
Q5 · The limits of the solution	Oliver Hartwell asks whether the firm can now answer the regulator's question about October. Give your honest answer and the boundary of what the solution can and cannot do.

LAB · FOUR DELIVERABLES, IN ORDER

The FCA wants a follow-up submission within thirty days: point-in-time attribution for every flagged position opened in Q3 2025, the full accountability chain, and evidence that roster history is now kept.

Setup Load	Load the two CSVs into `Warehouse_Veritas` as stg.Flagged_Positions and ref.Relationship_Managers. Land the staging columns as text. The positions file is Latin-1 encoded because of the £ sign.
D1 View	Build gold.Clean_Positions on stg.Flagged_Positions with defensive casting, a join to ref.Relationship_Managers that keeps positions with no matched RM, and an exposure band on every row. All Exhibit 06 columns. Check every D1 criterion before moving on.

D2 Leaderboard	Rank every RM by resolution performance from the view. Peers compete only within the same Title, only Resolved positions with sign-off count, and equal performance gives equal rank.
D3 Hierarchy	Produce the complete chain from ref.Relationship_Managers: RM_Name, Title, Hierarchy_Depth, Accountability_Chain. Recursive queries are not available in Fabric Warehouse. Every RM appears exactly once, and a missing level leaves no gap in the chain.
D4 SCD Type 2	Add history columns to ref.Relationship_Managers and backfill every existing row as active. Run the setup script below, then write one atomic MERGE that applies the changes in stg.New_RM_Roster: close the prior record, insert the new active one, never overwrite.

```
-- Setup for Deliverable 4: the incoming roster changes
CREATE TABLE [Warehouse_Veritas].[stg].[New_RM_Roster] (
    RMID INT, Name VARCHAR(100), Title VARCHAR(50),
    ManagerID INT, Jurisdiction VARCHAR(10), Desk VARCHAR(50)
);
INSERT INTO [Warehouse_Veritas].[stg].[New_RM_Roster]
    (RMID, Name, Title, ManagerID, Jurisdiction, Desk)
VALUES
    (101, 'Natalie Rousseau', 'Associate', 102, 'UK', 'Equities'),     -- promoted from Analyst
    (103, 'Wei Liang',        'Director',  104, 'UK', 'Multi-Asset'),  -- moved from SG to UK
    (106, 'Amir Hassan',      'Analyst',   102, 'US', 'Fixed Income'); -- new RM
```

AFTER DELIVERABLE 4

Query ref.Relationship_Managers and review the result with Priya Anand. Oliver Hartwell will ask whether the October question can now be answered; your answer must be honest and precisely bounded. Then rerun Deliverables 1 to 3 against the updated roster and confirm every criterion still holds.

CHALLENGES · AFTER THE ARCHITECTURE IS LIVE

Challenge 1 · Exposure concentration	Oliver Hartwell asks which desks and jurisdictions carry the highest concentration of unresolved critical risk: the share of total notional exposure in Critical Exposure positions still Open, by Desk and Jurisdiction, worst first. Write one query. The denominator is total notional across the whole portfolio, not the critical subset. Explain why that matters to a risk officer, why your ranking function beats its closest alternative, and how you bring one portfolio total into a grouped result without a subquery in the SELECT clause.
Challenge 2 · Hierarchy concentration	Priya Anand must brief each Managing Partner on the critical exposure anywhere below them, across all five levels. One row per Managing Partner: count and total notional of Critical Exposure Open positions, ranked by exposure. Write one query combining your hierarchy with position data. Explain which join type the two need and why, what your chain returns for a Managing Partner with no manager above, and why this query needs Deliverable 3.

If D1 to D4 are sound, both queries take a handful of lines. If not, neither resolves cleanly.

MICROSOFT FABRIC BOOTCAMP BY COE

From a roster that overwrites to a chain that survives audit.

A governed view that survives dirty source data, a leaderboard partitioned by title, a five-level hierarchy resolved without recursion, and an SCD Type 2 MERGE that keeps every roster change instead of overwriting it.

WHAT YOU DELIVER

Four T-SQL artefacts: gold.Clean_Positions, an RM leaderboard with DENSE_RANK by Title, a hierarchy flattener returning RM_Name, Title, Hierarchy_Depth and Accountability_Chain, and an SCD Type 2 MERGE. Plus the two challenge queries.

HOW YOU ARE EVALUATED

The seven acceptance criteria and five discussion answers framed against design intent.

WHAT THIS PROVES

You can write T-SQL that absorbs dirty data, rank deliberately, walk a self-referencing hierarchy without recursion, keep history with one atomic MERGE, and be honest about what the architecture can prove.

WHAT COMES NEXT

Monitor the count of sentinel dates daily. Alert when an RM's depth does not match their Title. Add row-level security on the view by jurisdiction.

INSTRUCTOR CONTACT

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